

Finding a Shorter Side

Pythagoras' Theorem. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

Sometimes you already know the longest side (the hypotenuse) and one of the shorter sides, and you need the other shorter side. This lesson builds a clear, step-by-step way to find it. The twist, compared with finding the hypotenuse, is that you **subtract** before you take the square root. The question you are exploring is: how can I use the hypotenuse and one shorter side to find the missing shorter side?

What you are learning to notice

- How to find the right angle and the hypotenuse (the longest side, opposite it).
- That the square on the hypotenuse is the largest — so here you take a square away from it.
- How to write $a^2 = c^2 - b^2$ for this triangle.
- That the square root, as always, is the final step that turns an area back into a length.

Moving through it, one step at a time

The builder walks through the steps: find the right angle and the hypotenuse; note which shorter side is missing; write the theorem for this triangle; square the hypotenuse and the known shorter side; subtract to find the missing square; then take the square root. Take them one at a time — each line follows from the one above.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (square numbers, square roots, and recognising that the hypotenuse carries the largest square), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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